Chuanyang Jin

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Education

New York University Sep 2020 – May 2024

B.A in Computer Science with Honors; B.A in Mathematics; received $18,000 in research funding *GPA: 3.96*

- Relevant coursework: Deep Learning (Grad), Machine Learning (Grad), Computer Vision (Grad), Natural Language Processing (Grad), Algorithms (TA), Robot Intelligence, Object Oriented Programming, Operating Systems, Computer Systems, Theory of Computation, Differential Equations, Numerical Analysis, Real Analysis

Publications / Preprints

- Kunal Jha, Tuan Anh Le, **Chuanyang Jin**, Yen-Ling Kuo, Joshua Tenenbaum and Tianmin Shu. Neural Amortized Inference for Nested Multi-agent Reasoning. Oral Presentation at AAAI Workshop 2023.
- **Chuanyang Jin**. Semi-Supervised Image Captioning with CLIP. Under Review.
- Jinjun Wu*, Yu Xiao*, Yunzheng Liu*, **Chuanyang Jin**, Xinqi Fan, Oded Regev, Chuan He, Jingyi Fei. The Composition of Nuclear Speckle and Relating Functions. Under Review.

Research Experience

Nested Reasoning May 2023 – June 2023

Summer Research Intern at MIT, Advisor: Prof. Josh Tenenbaum *Boston*

- Design and construct experimental scenarios within an I-POMDP framework to test nested multi-agent reasoning processes. Conduct experiments using baseline and our Neural Amortized Inference models. Co-author a paper.

Optimizing Diffusion Transformers Jan 2023 – present

Advisor: Prof. Saining Xie *New York*

- Implement features including pre-extracted VAE features, gradient checkpointing, and mixed precision training, achieving 95% speedup and 60% memory reduction, gain 120 GitHub Stars and integrated into official repository.
- Investigate 4-bit post training quantization of diffusion modes and progressive model distillation for fast sampling.

Interpretable Machine Learning for Biological Processes Jan 2022 – June 2023

Advisor: Prof. Oded Regev *New York*

- Present a neural network model that provides insights into RNA splicing.
- Use an "interpretable-by-design" approach, and achieve predictive accuracy on par with state-of-the-art models.
- Conduct extensive biological data analysis, trace and quantify the entire process from input to speckle enrichment prediction. The model reveals novel components of the splicing logic, which we experimentally validates.

Semi-Supervised Image Captioning with CLIP April 2023 – June 2023

- Propose a semi-supervised image captioning approach using CLIP encoding, mapping networks, and GPT, with CLIP image-text similarity as unsupervised training objective. Ablate several architecture and dataset choices.
- Outperform supervised method using 2% labeled data, measured by both reference-based and our novel metrics.

Work / Project Experience

Visual Place Recognition Using Perspective Images Jun 2022 – Aug 2022

NYU AI4CE Lab *New York*

- Enhance a visual place recognition model to accommodate perspective images along with panoramas, by leveraging an encoder-decoder model to perform scene completion. Finetune it via a pretext task of relative pose estimation.
- Conduct experiments on simulated and real datasets, outperforming baselines in recall rate and robustness.
- Develop a unified framework that handles various supervision methods, and both point cloud and RGB-D images.

Software Development Intern June 2021 – Aug 2021

Chinasoft International Ltd. *Nanjing*

- Maintain and extend the functionalities of a company's personnel/financial data management web platform.
- Design, implement, and test tools using MySQL to enable administrators to operate on shared databases.
- Leverage Django web framework, JavaScript and HTML/CSS to create a user-friendly interactive interface.

Efficient Video Prediction and Segmentation with Sparse Labeled Data April 2023 – May 2023

- Propose a multi-stage transfer learning algorithm along with a modified SimVP model to effectively address the challenge of scarce labeled data for end-to-end video prediction and segmentation tasks in a unified framework.
- Achieve a Jaccard Index of 0.403, ranking 3rd in 29 teams in a grad-level course supervised by Prof. Yann LeCun.

Class Embedding Patterns in Classifier-Free Diffusion Sep 2022 – Dec 2022

- Explore approaches to generate mappings from class labels and integrate them into the reverse diffusion process.
- Design and implement class label embedding patterns such as Uniform, Pyramid, and Bottleneck Embedding.
- Evaluate their performance, properties, and trade-offs. Leverage curriculum learning to handle noisy labels.

AutoCorp: A Fully Autonomous Company April 2023

- Build an AI-driven company by chaining LLMs, managing memory, creating interfaces, and automating decisions.
- After human gives initial direction, AutoCorp generates creative ads, branding, and product designs using GPT-4/DALL-E/ControlNet. Automatic prompt updates are triggered by customer events and feedback.

Parallel Command Processor Sep 2022 – Dec 2022

- Build a command interpreter that handles I/O redirection and pipe, and manages background/foreground process.
- Implement a thread pool using POSIX threading to parallelize provided tasks for improved performance.

Graph Visualization of Matrices March 2022 – May 2022

- Introduce a novel shift method to speed up the convergence of power iterations for computing tensor eigenpairs.
- Implement dimension reduction with power iterations and propose an innovative *graph Laplacian* concept for matrix visualization. Leverage them to create a toolbox that helps uncover structures in complex real-world relations.

Promoting Global Equity through Asteroid Mining Feb 2022 – March 2022

- Project a vision for asteroid mining and its potential impact on global equity through analytic hierarchy processes.
- Detail a profit distribution scheme that considers fairness-efficiency trade-off using PDE-constrained optimization.

Gene Mutation Detection Platform May 2019 – Aug 2021

- Propose a mutation detection algorithm based on computational geometry and parameterized by deep learning.
- Launch an online platform adopted by leading hospitals to reduce diagnosis periods and receive a $10k reward.

Honors/Awards

- Computer Science Prize for Most Promising Student (Top 1) in the Junior Year at NYU 2023
- Finalist & MAA Award & COMAP Scholarship (Top 4/27205), Interdisciplinary Contest in Modeling 2022
- Champion (Top 1/51) of Tsinghua X-lab AI Hackathon 2021
- Distinguished Honor Roll (Top 1%) in AMC12 & USAMO qualifier 2020
- International Bronze Award (Top 3/500), Shing-Tung Yau High School Computer Science Award 2019
- Finalist (Top 8/3790) of FIRST Robotics Competition World Championship 2019

Services/Activities

- Teaching Assistant of CSCI-UA 310 Basic Algorithms (Fall 2022)
- Executive board member of NYU Mathematics Society (2021 – 2023)
- Director of Environmental Sustainability Committee at NYU Shanghai (2020 – 2021)
- Influential writer on Zhihu Q&A forum with 26K followers and 15,000K total reads

Skills

- Language: Python, C++, C, Java, JavaScript, Matlab, Julia, SQL, R, Octave, Linux, Git
- Framework: PyTorch, TensorFlow, Scikit-Learn, JAX, Transformers, OpenCV, OpenMP